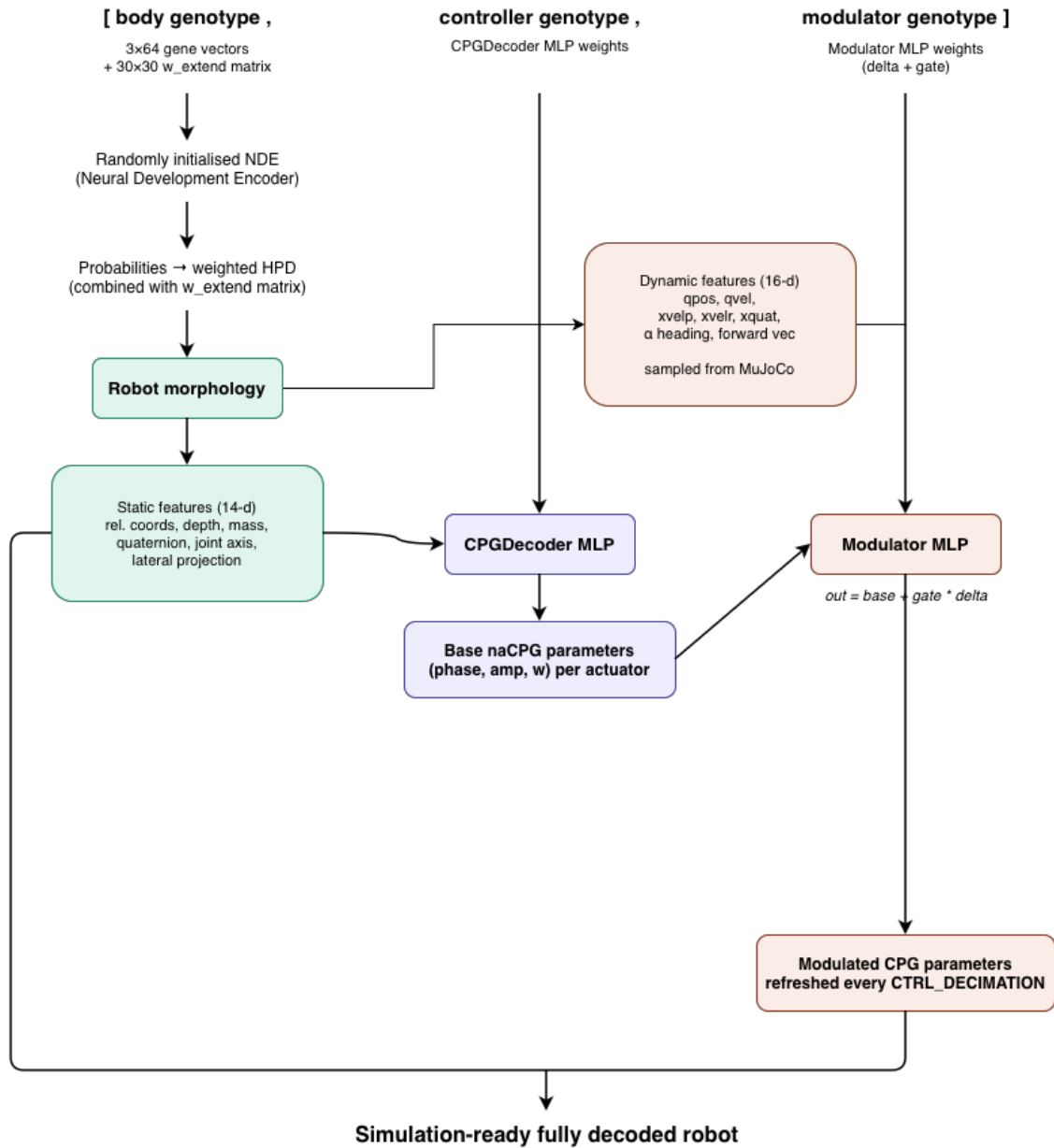


DECODING



Static features — 14-d per actuator Computed once at decode, fed to CPGDecoder.

- **rel_pos (3)** — joint body's (x, y, z) position relative to the core. Tells the decoder *where in space* this joint sits on the robot.
- **euclidean_dist (1)** — distance from joint to core. Compact summary of "how far out" this limb is.
- **depth (1)** — depth in the morphology graph (shortest-path hops from core). Captures topology, not geometry — a far-flung leaf vs. a close-by limb.
- **xquat (4)** — body orientation quaternion at decode time. The joint's rest-pose attitude relative to world.
- **subtreemass (1)** — total mass of this body's subtree. Lets the decoder differentiate a heavy leg from a light flap.
- **joint_axis (3)** — unit vector of the joint's rotation axis. Distinguishes pitch joints from yaw/roll joints.
- **lateral_proj (1)** — scalar projection of rel_pos[:2] onto the robot-frame lateral axis. Signed left/right offset from the body midline.

Dynamic features — 16-d per actuator *Resampled from live MuJoCo state every CTRL_DECIMATION steps, fed to Modulator.*

- **qpos (1)** — current joint angle. The CPG's actual instantaneous position.
- **qvel (1)** — current joint angular velocity. How fast it's moving right now.
- **xvelp (3)** — linear velocity of this body in world frame. Captures how this limb is translating through space.
- **xvelr (3)** — angular velocity of this body in world frame. Captures how this limb is rotating.
- **xquat (4)** — *current* body orientation quaternion. Same shape as the static one but live (the limb may have rolled, pitched, etc. during locomotion).
- **alpha_arr (2)** — $[\sin(\alpha), \cos(\alpha)]$ of the heading error, where α = target direction – current heading. Encoded as sin/cos to avoid the $-\pi/+\pi$ wrap discontinuity. Same value across all actuators (global state, replicated).
- **forward_vec (2)** — current (x, y) forward direction of the core in world frame. Also global, replicated across rows. Together with alpha, this is what lets the modulator implement *targeted* locomotion — it tells each joint where the robot is pointing vs. where it should be going.

Hyperparameters

Simulation

- $dt = 0.01$ — MuJoCo physics timestep
- $steps = 1800$ — total physics steps per roll-out
- $BUFFER_FALL = 300$ — settle window before control + logging start (was 350)
- $CTRL_DECIMATION = 70$ — modulator refresh interval (was 100)
- $CONTACT_CHECK_START = 300$ (= $BUFFER_FALL$)
- $CONTACT_CHECK_END = 600$ (was 700)
- $MAX_EARLY_CONTACTS = 1500$ — cumulative ncon threshold; exceeding it $\rightarrow$ fitness = $-\infty$ (was 1700)

Morphology / decoding

- $NMODULES = 30$ — max modules per robot
- $GENE_SIZE_NDE = 64$ — per-channel gene-vector length ($\times 3$ for type/conn/rot = 192)
- w_extend matrix size = $30 \times 30 = 900$ per individual
- Body gene init: $U(-2.0, 2.0)$
- Controller + modulator gene init: $U(-1.0, 1.0)$

Fitness

- $target_directions_deg = [0, 120, 240]$ — 3 directions averaged in $performance_3p_tl$
- $fitness_function_w = 0.1$ — lateral-error penalty weight
- $\epsilon = 1e-9$ — numerical guard

EA — RevDE

- $POP_SIZE = 64$
- $N_GENERATIONS = 60$
- $F = \{body: 0.3, w_extend: 0.3, controller: 0.8, modulator: 0.8\}$ — per-block DE scaling
- $P = 0.5$ — DE binary-mask crossover probability

EA — selection (l_{delta})